

Suricata Setup

Installation and integration of Suricata on the SOC_Lab_Endpoint agent

1. Installation of Suricata

First, we add the official repository and install

```
sudo add-apt-repository ppa:oisf/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y
```

Verify the installation and version

```
suricata -V
```

2. Install Curl

Install curl to allow the system to fetch the Emerging Threats ruleset.

```
sudo apt install curl -y
```

3. Download and Install Rules

To detect threats, Suricata needs rules, download the Emerging Threats (ET) Open ruleset

```
cd /tmp/
curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz
sudo mkdir -p /etc/suricata/rules
sudo mv rules/*.rules /etc/suricata/rules/
```

4. Configure suricata.yaml

Configure Suricata to recognize our local network and point to the correct rules directory.

```
sudo nano /etc/suricata/suricata.yaml
```

Key modifications to include in the file:

- i. Ensure %YAML 1.1 is at the very top of the file
- ii. **Network Variables:**
Home network -Set this to your local network range
HOME_NET: "<SOC_Lab_Endpoint IP>"
EXTERNAL_NET: "any"

```
Terminal May 8 10:01 osboxes@osboxes: ~
GNU nano 6.2 /etc/suricata/suricata.yaml *
%YAML 1.1
---
# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html

# This configuration file was generated by Suricata 8.0.3.
suricata-version: "8.0"

##
## Step 1: Inform Suricata about your network
##

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    EXTERNAL_NET: "any"

    HTTP_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
    SQL_SERVERS: "$HOME_NET"
    DNS_SERVERS: "$HOME_NET"
    TELNET_SERVERS: "$HOME_NET"
    AIM_SERVERS: "$EXTERNAL_NET"
    DC_SERVERS: "$HOME_NET"
    DNP3_SERVER: "$HOME_NET"
    DNP3_CLIENT: "$HOME_NET"
    MODBUS_CLIENT: "$HOME_NET"
```

iii. Rule Path & Files — Ensure rules load

```
default-rule-path: /etc/suricata/rules
rule-files: - "*.rules"
            - "*.rules"
```

```
GNU nano 6.2 /etc/suricata/suricata.yaml *
# When auto-config is enabled the hashmode specifies the algorithm for
# determining to which stream a given packet is to be delivered.
# This can be any valid Napatech NTPL hashmode command.
#
# The most common hashmode commands are: hash2tuple, hash2tuplesorted,
# hash5tuple, hash5tuplesorted and roundrobin.
#
# See Napatech NTPL documentation other hashmodes and details on their use.
#
# This parameter has no effect if auto-config is disabled.
#
hashmode: hash5tuplesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /etc/suricata/rules
rule-files:
  - "*.rules"

##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Suricata as a Firewall options (experimental)
```

iv. Get the interface of your ubuntu

Ip a

```
osboxes@osboxes: ~  
osboxes@osboxes:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5f:8e:96 brd ff:ff:ff:ff:ff:ff
```

v. Packet Capture Interface

IP a output must be the same as packet capture interface

```
af-packet:  
- interface: enp0s3
```

```
GNU nano 6.2 /etc/suricata/suricata.yaml *  
# format: "[%i - %m] %Z %d: %S: %M"  
# type: json  
- syslog:  
    enabled: no  
    facility: local5  
    format: "[%i] <%d> -- "  
    # type: json  
  
##  
## Step 3: Configure common capture settings  
##  
## See "Advanced Capture Options" below for more options, including Netmap  
## and PF_RING.  
##  
  
# Linux high-speed-capture support  
af-packet:  
- interface: enp0s3  
  # number of receive threads. "auto" uses the number of cores  
  #threads: auto  
  # Default clusterid. AF_PACKET will load balance packets based on flow.  
  cluster-id: 99  
  # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per  
  # This is only supported for Linux kernel > 3.1  
  # possible value are:  
  # * cluster_flow: all packets of a given flow are sent to the same socket  
  # * cluster_cpu: all packets treated in kernel by a CPU are sent to the sa  
  # * cluster_qm: all packets linked by network card to a RSS queue are sent  
  # socket. Requires at least Linux 3.14.
```

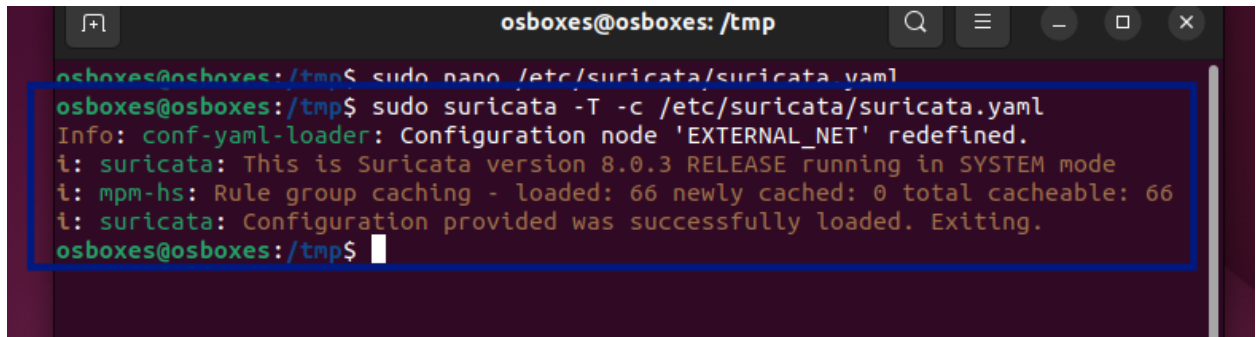
Save the configuration file with **Ctrl O**

Press **enter**

Press **Ctrl X** to exit

To verify the configuration changes

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

A terminal window titled 'osboxes@osboxes: /tmp' showing the execution of the command 'sudo suricata -T -c /etc/suricata/suricata.yaml'. The output includes: 'Info: conf-yaml-loader: Configuration node 'EXTERNAL_NET' redefined.', 'i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode', 'i: mpm-hs: Rule group caching - loaded: 66 newly cached: 0 total cacheable: 66', 'i: suricata: Configuration provided was successfully loaded. Exiting.', and the prompt 'osboxes@osboxes: /tmp\$'.

5. Restart Suricata:

```
sudo systemctl restart suricata  
sudo systemctl status suricata
```

Press Ctrl C or just Q to exit

6. Integrating with Wazuh Agent

To send Suricata's logs to Wazuh Manager, edit the Wazuh Agent's configuration file on the SOC_Lab_Endpoint

```
sudo nano /var/ossec/etc/ossec.conf
```

Add the following block to the **<ossec_config>** section so Wazuh can read Suricata's JSON output:

```
<localfile>  
  <log_format>json</log_format>  
  <location>/var/log/suricata/eve.json</location>  
</localfile>
```

```
osboxes@osboxes: /tmp
GNU nano 6.2 /var/ossec/etc/ossec.conf
<active-response>
  <disabled>no</disabled>
  <ca_store>etc/wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active-response>

<!-- Choose between "plain", "json", or "plain,json" for the format of intern
<logging>
  <log_format>plain</log_format>
</logging>
</ossec_conf>

<ossec_conf>
  <localfile>
    <log_format>journald</log_format>
    <location>journald</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>
</ossec_conf>
```

7. Restart Wazuh Agent:

```
sudo systemctl restart wazuh-agent
```

8. Testing and Verification

i. Generate Traffic

Run a simple ping test to simulate network activity:

Use the SOC_Lab_Endpoint IP

```
ping -c 20 <UBUNTU_IP>
```

```
Session Actions Edit View Help
kali)~$ ping -c 20 192.168.1.8
PING 192.168.1.8 (192.168.1.8) 56(84) bytes of data.
64 bytes from 192.168.1.8: icmp_seq=1 ttl=64 time=1.06 ms
64 bytes from 192.168.1.8: icmp_seq=2 ttl=64 time=0.898 ms
64 bytes from 192.168.1.8: icmp_seq=3 ttl=64 time=1.43 ms
64 bytes from 192.168.1.8: icmp_seq=4 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=5 ttl=64 time=1.10 ms
64 bytes from 192.168.1.8: icmp_seq=6 ttl=64 time=1.35 ms
64 bytes from 192.168.1.8: icmp_seq=7 ttl=64 time=0.977 ms
64 bytes from 192.168.1.8: icmp_seq=8 ttl=64 time=1.77 ms
64 bytes from 192.168.1.8: icmp_seq=9 ttl=64 time=1.37 ms
64 bytes from 192.168.1.8: icmp_seq=10 ttl=64 time=0.923 ms
64 bytes from 192.168.1.8: icmp_seq=11 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.1.8: icmp_seq=13 ttl=64 time=1.31 ms
64 bytes from 192.168.1.8: icmp_seq=14 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=15 ttl=64 time=1.46 ms
64 bytes from 192.168.1.8: icmp_seq=16 ttl=64 time=1.18 ms
64 bytes from 192.168.1.8: icmp_seq=17 ttl=64 time=1.36 ms
64 bytes from 192.168.1.8: icmp_seq=18 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=19 ttl=64 time=1.17 ms
64 bytes from 192.168.1.8: icmp_seq=20 ttl=64 time=0.955 ms
— 192.168.1.8 ping statistics —
20 packets transmitted, 20 received, 0% packet loss, time 19226ms
rtt min/avg/max/mdev = 0.898/1.232/1.771/0.218 ms
```

ii. Verify Alerts

Navigate to the Wazuh Dashboard and check Threat Hunting.

